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SUMMARY OF QUALIFICATIONS

Results-driven creator of decision-making software solutions for factory operations. Organization leader that excels at recognizing opportunities for improvement and solving complex problems in a fast-paced environment.

Technical leader for multiple developer teams to establish high level strategies, standards, frameworks, and architecture models for enterprise-wide systems to leverage modern web applications. Program manager capable of coordinating efforts from many globally distributed teams as well as mentoring and training other developers, architects and business partners on reliable and scalable best known methods.

Skills emphasis in analyzing and streamlining many factors of operations in order to increase productivity, quality, and efficiency. Expert developer experienced in all software engineering roles including front-end, back-end, test, DevOps, Security, and Cloud infrastructure. Specializing in full-stack solutions involving data collection, analysis , and visualization in support of a variety of semi-conductor factory operations.

WORK EXPERIENCE

IT Enterprise Architect (2019 - present) – Micron Technology

- Organized the Enterprise Web Application road-map for Micron and lead a global effort to align solutions to newly defined standards. Established the O2 framework (originally Omelek) as the base road-map for most all new solution development at Micron.
- Built out infrastructure for the shared framework to host all Micron web applications in cloud-based environments both on-premise leveraging OpenShift and off-premise with Azure. Enabled complex data-level security rules to be applied from distributed sources (data systems and source API's) throughout the network.
- Established a suite of front-end software libraries written in Angular and back-end & middle-ware tools written in .NET Core to provide end-users with a consistent experience and developers with many necessary Micron customizations built in and code/component re-use to accelerate development cycles.
- As the leader of a large community of developers enabled migration of hundreds of legacy stand-alone (executable) solutions to a new modern web framework through multiple trainings and presentations.
- Enabled platform and developers to create and sustain over 200 different complex decision-making solutions and a developer community of over 900 team-members. Establishment of aligned platform has saved Micron millions in cost-savings through consolidation strategy and cost avoidance with provided capabilities.

Principal Operations Performance Engineer (2016 - 2019) – IM Flash Tech - Data Technology

- Omelek – Participated in the foundational governance and development of the core library for web applications at Micron and enabled early initial global adoption. Implemented many code changes that were key enablers for adoption by a wide variety of user teams as a non-IT team member. Led several training presentations and week-long events to help introduce new developers to the platform and gain support in alignment and adoption. Helped to create documentation for development teams to leverage a wide variety of features from within the platform as well has how to get started in a shared instance leading to a large initial development community.
- xWeb Platform – Helped Micron central team establish one of the first Omelek solutions named "xWeb" with a model for global open source contribution. Enabled various migrations and uplifts to the instance including new Angular versions, the migration to the OpenShift Cloud environment, and enablement of a CI/CD solution for automated deployments open to all coders (including citizen-developers). Adoption has grown in usage to be one of the most heavily used web applications in the network with over 150 applications. Helped establish the initial documentation for creating a widget library in the xWeb instance and content has been kept up and used in almost all trainings since inception.
- xWeb Applications – I was the primary developer for over 10 applications running within xWeb today and am the primary for the top 3 most heavily used widgets. Some of the key widgets are "Tickets", "eToolmon", "Corrective Action Feed", "SWR Report". I also created a "fab-common"

library for sharing several pure components and functions to users which is used by many widgets in the xWeb instance.

- Data Engineering – I also have led the team in organizing several key data infrastructure environments including MSSQL database administration and both MongoDB and CouchDB for NoSQL document stores. This in combination with the development of several API layer solutions has been key to the success for several integrated solutions. I have created API solutions to integrate these data layers using several languages including Node.js, .NET Core, and Python.
- Team Leadership – Organized and led a team product catalog effort to review and evaluate all products supported by the Operations Performance team. Helped the team establish peers for ongoing work on all key solutions and established a health score matrix to evaluate which solutions required the most immediate attention for support and uplift.
- Helped the initial Lehi Operations Performance team establish and migrate to an aligned web development road-map from a starting point of various legacy technologies. Helped to establish standards and best-known methods leveraging the new road-map for a consolidated set of key technologies to enable alignment and future growth.

Sr. Operations Improvement Engineer (Sept 2007 – 2016) – IM Flash Tech - Industrial Engineering

- Scheduler deployment – Lead the effort to establish the first fully-automated scheduling system know as GHOST as the area scheduling solution for Micron Facilities world-wide. This was accomplished via a proof of concept R&D phase for several key areas beginning with Photo. The success seen via deployment in the F2 Photo area demonstrated the potential as several key performance metrics achieved record levels. The solution now has since been adopted by all areas in the Fab and has also been successfully shared and deployed to multiple sites as the current benchmark solution.
- WW Training & Collaboration – Leader and organizer of the first scheduling global collaboration meetings which transitioned to be led by a global rotation of technical leaders. The collaboration has meeting time has become a key element of growth of a global team for continuous improvement in all facets of area scheduling.
- Line Priority Solution **–** As a key contributor to a world-wide team we have recognized and created a road-map to rework the current line priority methods used in our scheduling system. I was a key leader in that team helping to organize a systematic method (static simulation model) of evaluating solution alternatives.
- Command Center & Remote Operations – Worked with team of production counterparts to design, create, and deploy command center workstations in the Fab. These workstations incorporate a variety of KPI monitoring systems customized to the needs of each individual manufacturing technician’s roles. These tools (software) at the PC stations in the Fab enable remote monitor and troubleshooting to quickly identify potential problems and opportunities for improvement and increased overall productivity.
- AutoStaging – Completed initial AutoStage deployment at Fab 2 along with many logic optimizations and Fab metric establishment to track and find opportunities and need for continuous improvement.
- Involved in early integration and development of factory automation for “AutoStage” and “AutoQual” systems and debugging.

Planning Engineer (Mar. 2005 – Sept 2007) – Micron Tech – Industrial Engineering Capacity Planning

- Fab 6 (Manassas 300mm) – Initial design and implementation of Idle State Visibility reporting to provide real time feedback to production of equipment status and potential lot issues.
- Fab 6 Auto-staging rules – Developed remaining auto-staging rules for Wet hoods and Diffusion. Fab 6 Training – Provided training to team member developing their skills with APF / RTD / Activity manager software used for reporting, debugging, and scheduler rule authoring.
- Fab 3 (Boise 200mm) Workstation Photo Strategy Report – Report designed to help production leadership making tool changeover decisions based on WIP an incoming product.
- Fab 3 Smart Sharing – Developed a report to help improve unplanned shared capacity – WIP building transfer decisions. Report will suggest transport if it will reduce cycle time.
- Fab 3 LDI/RTD rules – Owned scheduling rules and code used to prioritize product movement throughout the site. Maintain common code for which is shared among multiple Micron sites.

Operator (Dec. 2004 - Mar. 2005) – Micron Tech - Diffusion Production

- Obtained Operator 2 certification and shared lead responsibilities

Calibration Service Manager – (2002-2003) - Branom Instrument Company, Seattle, WA

- Facilitated significant cost savings by developing organized system for administrative management of product flow of calibration service department.
- Designed and administered database used to collect instrument calibration data and generate certification reports to meet tractability requirements of NIST and ANSI/Z540.
- Reduced order lead-time for service department from average of over two weeks down to an average of two or three days.

University of Washington College of Engineering, Seattle, WA
Dean’s Office Computing Support Technician (1998-2002)

- Ground up design of automated system for authorizing and selling software licenses
- Responsible for setup, repair, and support of computers within dean’s office.

EDUCATION

B. S., Industrial/Manufacturing Engineering - University of Washington, Seattle, WA, 2002.

- Recipient of Washington Pulp and Paper Foundation (WPPF) Scholarship.
- Member of National Institute of Industrial Engineers (IIE) and Society of Automotive Engineers (SAE)

SUMMARY OF EXPERTISE

Engineering / Business

- Statistics, Design of Experiments, Reliability, Statistical Quality Control, Six Sigma
- Manufacturing Scheduling, Linear Programming, Optimization, MRP
- Manufacturing Simulation (Arena, Autosched, AutoMod)
- User interface design, Ergonomics, and Human factors
- Engineering Economics, Macroeconomics, Microeconomics
- Mechanics of Materials, Applied Physics, Statics, Dynamics
- Project Management, Electrical Engineering, Computer Science, Data Visualization

Technical / Computing

- Software Expertise — Windows, Office, Linux, JMP, AutoMod, GIT, Cloud, Docker, Kubernetes, OpenShift, Azure, AWS
- Web Languages — HTML, JavaScript, TypeScript, PHP, D3, HighCharts, Plotly, AJAX, jQuery Angular, React, Vue, Bootstrap, RXJS, NGRX, Redux
- Other Programming — C# .NET, Python, R lang, Node.js, Shell scripting
- Database — AMAT APF, MS SQL, Oracle, Snowflake, Hbase, Hive, Postgres, MySQL, CouchDB, MongoDB